An Exact Fold Condition for Mass-Balanced Models of Protein Quality Control

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Abstract

Supplementary material accompanying the paper above. Every number in the caption is recomputed by the script named in it and asserted by the accompanying test suite.

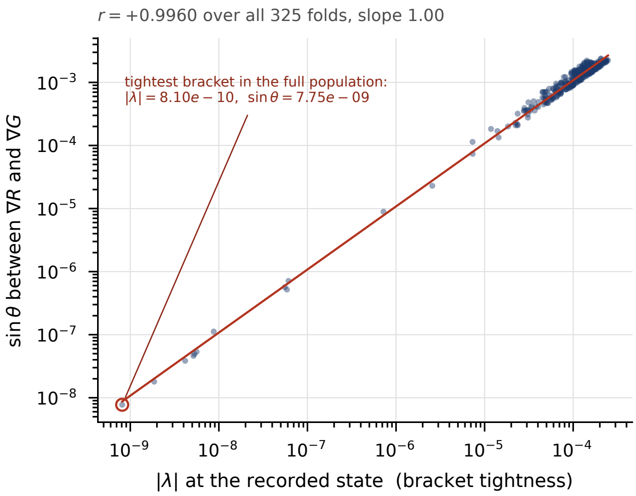

Fig. S1 Parallelism residual against leading eigenvalue, over all 325 load-grid folds with no subsample. The correlation is +0.9960 in log-log with slope 1.000, so sin($angle$) is proportional to bracket looseness rather than merely increasing with it; the tightest-bracketed state has $|\lambda| = 8.10{\times}10^{-10}$ and sin($angle$) $= 7.75{\times}10^{-9}$. The figure also carries the normalisation contrast: normalising by $\max(|\det J|, |\,\mathrm{cross}\,|)$ is 0/0 at an exact fold, and that residual degrades as the bracket tightens (median $3.133{\times}10^{-7}$ on the tighter half against $1.455{\times}10^{-7}$ on the looser, 2.15× worse where it matters most, log-log correlation −0.835, maximum $1.54{\times}10^{-2}$), whereas the gradient normalisation used throughout correlates at +0.041 with a maximum of $1.29{\times}10^{-9}$.

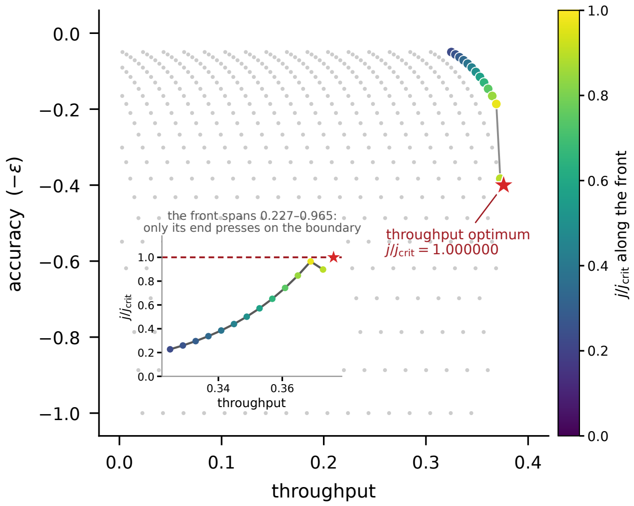

Fig. S2 Strategy space cut by the fold condition: 469 feasible strategies, 13 non-dominated, with j/j_{crit} running 0.2271 to 0.9652 along the front (median 0.5010). The exact throughput optimum sits at j/j_{crit} = 1.000000 exactly, on the feasibility boundary; the best grid strategy reaches 0.897488, the gap being discretisation of the grid rather than an interior optimum. Supplementary Section S1 develops a strategy-space consequence of the fold condition: cutting a two-coordinate trade-off surface by the constraint $j(\alpha, R) < j_{crit}(R)$ places the throughput optimum exactly on the feasibility boundary.